

Maternal Health in Sub-Saharan Africa Betta Edu

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Abstract

Maternal health refers to the health of women during pregnancy, childbirth, and the postpartum period, and this definition conforms to the attainment of Sustainable Development Goal three (SDG 3). However, achieving this SGD remains a mirage in sub-Saharan Africa, as the region continues to possess the highest rates of maternal mortality across the globe. Large intraregional disparities in terms of coverage of basic maternal health still exist across the region, and the causes of these preventable deaths must be analyzed frequently to guide interventions if progress is to be made. The health status of pregnant women remains an important indicator of the general state of health of any given population and also influences the health of their babies. Hence, fetal and infant health is one of the main determinants of health in childhood and adulthood. Maternal health still faces multilayered challenges like inadequate infrastructure, low education, and financial constraints, yet also holds significant prospects for improvement through targeted interventions and policy changes. This chapter explores maternal health, contextualizing and localizing the sub-Saharan African perspectives, taking into consideration the region's challenges and prospects for improving and mitigation strategies toward maternal health in sub-Saharan Africa, focusing on the impact of maternal health on both women and their children, and the need for a more holistic approach to improve services and address underlying social determinants.

Keywords: maternal health, sub-Saharan Africa, challenges, prospects, mitigation

1. Introduction to maternal health

Maternal health remains a primary public health concern globally. Maternal health refers to the health of women during pregnancy, childbirth, and the postnatal period, ensuring women and their babies reach their full potential for health and well-being [1, 2]. Each stage presents a series of important events and changes, including physical, emotional, and social, occurring before, during, and well after the 40 weeks of gestation and the first year after childbirth [3]. These changes, although natural, can be stressful, and they have health consequences across a woman's life course. Maternal mortality, as defined by the Bureau of Vital Statistics, is the death of a woman while she is pregnant or within 42 days of termination of pregnancy, despite duration and site of pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from accidental or incidental causes [4]. The situation is exacerbated in sub-Saharan Africa (SSA), where the region accounts for a disproportionate share of global maternal deaths due to inequalities in access to quality health care services, weak health systems, and poor policies and guidelines of obstetric care [5, 6].

The reduction of maternal mortality and the promotion of maternal health and well-being are complex tasks. In recent years, the focus of maternal and child health efforts has gradually shifted from reducing maternal and infant mortality to pursuing high-quality maternal and child health care, against the backdrop of progressive improvements in global health care and increasing national attention to maternal and child populations [2, 7]. However, this has not translated to improving the maternal and child health indices in developing countries. In 2020, the Maternal Mortality Ratio (MMR) in sub-Saharan Africa was 536 per 100,000 live births, accounting for about 70% of all maternal deaths globally [8]. This suggests that the lifetime risk of maternal death in sub-Saharan Africa is 1 in 41, which is 268 times higher than in Western Europe. Although substantial progress has been made over the past two decades to increase access to and use of higher-quality healthcare during pregnancy and childbirth to prevent deaths and diseases, as well as improve women and adolescent girls' experience of pregnancy and childbirth, the indicator of the Sustainable Development Goal three (SDG 3), which is aimed at improving the proportion of births

attended by skilled health personnel (doctor, nurse, midwife), is almost becoming unrealistic [5, 6].

Despite overwhelming evidence from literature on the positive effect of maternal health care services in mitigating the risk of deaths due to pregnancy and childbirth complications, most developing countries continue to record low access to these services [9]. Facility-based delivery has been proven to be an essential factor in preventing maternal death. Yet, a significant number of women in sub-Saharan Africa still deliver at home without health professionals' assistance [10]. There are still disparities in the region, and factors determining these discrepancies are sparse. Therefore, maternal health constitutes a high-priority agenda for governments across the sub-Saharan region.

2. Challenges relating to maternal health in sub-Saharan Africa

There are numerous challenges relating to maternal health, and these challenges are multidimensional in nature. Empirical evidence suggests that the SSA remains one of the regions with modest health outcomes, as evidenced by high maternal mortality ratios and under-five mortality rates [11]. There are complications that occur during and following pregnancy and childbirth that can contribute to maternal deaths, most of which are preventable or treatable. First, evidence shows that early and regular attendance of antenatal care and delivery in a health facility under the supervision of trained personnel is associated with improved maternal health outcomes [12]. This may not be unrelated to limited healthcare financing. In sub-Saharan Africa (SSA), health spending per capita is significantly lower than in other regions, with countries spending an average of about \$92 per person in 2021 [13]. This is considerably less than, for example, the \$379 per person spent in the North Africa and Middle East region [14]. The health budget in SSA often faces challenges due to limited government spending, donor dependence, and disparities in health outcomes across the region. This has also led to a critical shortage of healthcare providers, particularly midwives and obstetricians [13, 15]. Secondly, geographical barriers, coupled with traditional practices and cultural beliefs, have also moderated access to health care [16, 17]. Many women in SSA live in remote areas, some of which are hard-to-reach settings and underserved communities, making it difficult to access healthcare services [18, 19]. Thirdly, conflict and instability in some SSA countries have disrupted healthcare services, therefore intensifying maternal health challenges, which have additionally induced poor healthcare infrastructure, lack of basic equipment, supplies, and other deliverables [20-22]. Moreover, poverty and inadequate female education have also contributed to the high rates of teenage pregnancies as well as the risk of maternal complications [23, 24]. Also, the issue is bordering on the high prevalence of the HIV/AIDS epidemic, which has contributed to the high MMR in SSA [25].

3. Prospects for improving and mitigation strategies toward maternal health in sub-Saharan Africa

Sub-Saharan Africa faces significant challenges in improving maternal health, but it also holds promising prospects for improvement. Even though many countries in the region subscribed to the United Nations Sustainable Development Goals, achieving these goals necessitates a deliberate collaborative effort between states, countries, regional governments, and international communities to establish a comprehensive framework for implementing solution-based strategies to address maternal and child health on a population scale [26].

3.1 Prospects for improving maternal health in sub-Saharan Africa

3.1.1 Progress in reducing maternal mortality

While high, maternal mortality rates in SSA have shown a decline, indicating that improvements are possible. Reducing maternal mortality in sub-Saharan Africa requires a multipronged approach, focusing on improving access to quality healthcare, addressing social determinants, and strengthening health

systems [27]. Key prospects include increasing access to skilled birth attendance, improving emergency obstetric care, promoting antenatal and postnatal care, and addressing social and economic factors that contribute to maternal mortality [8, 28].

3.1.2 Technological advancements

Innovations in maternal health, such as digital solutions and community health worker programs, offer potential for reaching more women and improving care [29]. Technological advancements offer significant prospects for reducing maternal mortality in sub-Saharan Africa. Telemedicine, mobile health apps, and AI-powered diagnostic tools can enhance access to care, improve the quality of services, and support remote monitoring [30, 31]. These technologies can also empower women with knowledge through health education initiatives, ultimately leading to better health outcomes.

3.1.3 Focus on emergency obstetric care

Strengthening emergency obstetric care (EmOC) capabilities is crucial for saving lives, particularly in rural areas. Focusing on emergency obstetric care (EmOC) is a key prospect for reducing maternal mortality in sub-Saharan Africa [32]. By ensuring access to skilled health personnel and timely interventions, especially for complications like obstetric hemorrhage and hypertensive disorders, EmOC can significantly improve maternal health outcomes [32, 33].

3.1.4 Investing in women's empowerment

Improving women's education and autonomy can lead to increased access to healthcare and better health outcomes. Investing in women's empowerment in sub-Saharan Africa holds strong prospects for reducing maternal mortality [34]. Empowered women are more likely to access quality healthcare, including antenatal and postnatal care, and are better able to make informed decisions about their reproductive health [35, 36]. This can lead to fewer complications during pregnancy and childbirth, ultimately lowering maternal mortality rates.

3.1.5 Male involvement; a prospect for improving maternal health in sub-Saharan Africa

Involving men in maternal healthcare can improve women's access to services and promote healthier practices [37]. Engaging men in maternal health in sub-Saharan Africa holds strong prospects for reducing maternal mortality by addressing gender inequalities and promoting better care-seeking behaviors [38]. By actively involving men in antenatal care, childbirth, and postpartum support, interventions can improve women's access to services, empower them in decision-making, and ultimately lead to safer pregnancies and childbirths [39].

3.1.6 Sustainable Development Goal (SDG) targets

The global push to reduce maternal mortality through the SDGs provides a framework for action and accountability [40]. The prospects for reducing maternal mortality in sub-Saharan Africa through the Sustainable Development Goal (SDG) targets are promising but require sustained and accelerated efforts [41]. While the region currently faces high maternal mortality rates, the SDG framework, particularly target 3.1, sets a clear goal of reducing the global maternal mortality ratio to less than 70 per 100,000 live births by 2030 [42].

3.2 Mitigation strategies for addressing issues bordering on maternal health in sub-Saharan Africa

3.2.1 Improve access to quality healthcare

This includes strengthening health systems, ensuring access to skilled birth attendants, and providing quality antenatal and postnatal care.

3.2.2 Address social determinants of health

Addressing poverty, lack of education, and other social factors that impact women's health is crucial.

3.2.3 Empower women

This includes providing women with access to education, economic opportunities, and decision-making power.

3.2.4 Increase men's involvement

Encourage men to actively participate in maternal healthcare, including attending antenatal care appointments, supporting their partners, and participating in decision-making.

3.2.5 Promote community-based interventions

Community health workers and other community-based programs can play a vital role in reaching women in remote areas and providing essential services.

3.2.6 Strengthen data collection and monitoring

Accurate data is essential for tracking progress and identifying areas for improvement.

3.2.7 Invest in research and development

Continued research is needed to identify innovative solutions for improving maternal health.

4. Conclusion

Maternal mortality is still persisting at an unacceptably high rate in the SSA region, with a variety of complex factors challenging sustained progress. However, key gray areas of focus could include, but not be limited to, increasing access to healthcare; improving healthcare infrastructure, especially in rural and underserved communities; training healthcare providers, particularly midwives and obstetricians; promoting family planning and reproductive health services to reduce teenage pregnancies; engaging with communities to address cultural barriers and promote healthcare-seeking behaviors; and strengthening healthcare financing systems to ensure that women can access healthcare services without financial hardship. By addressing these hydra-headed challenges and implementing interventions that are context-specific and incorporating the human-rights-based approach to maternal and child health intervention, sub-Saharan African countries can reduce maternal mortality and improve maternal health outcomes.

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